Customer Behavior Report Analysis

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Customer Shopping Behavior Analysis

1. Project Overview

This project focuses on analyzing customer behavior data to extract meaningful business insights. The dataset was provided in CSV format and was cleaned using Python (Pandas). After handling missing values and performing exploratory data analysis, key patterns were identified. An interactive Power BI dashboard was created to visualize revenue trends, customer segments, and subscription performance.

2.Dataset Summery :

The dataset used in this project was provided in CSV format and contains customer transaction and behavioral data. It includes information such as customer demographics, purchase details, subscription status, product categories, shipping types, and review ratings.

row : 3900
columns : 19
total missing values : 37
Keyfeatures : Customer ID , Age , Gender , Item Purchased , Category , Purchase Amount (USD) , Location , Size , Color , Season , Subscription Status , Shipping Type , Discount Applied , Promo Code Used , Previous Purchases , Payment Method , Frequency of Purchases

3.Exploratory Data Analysis using Python

After completing the data cleaning and preprocessing steps, Exploratory Data Analysis (EDA) was performed to understand customer purchasing behavior and identify meaningful patterns within the dataset using SQL.

1.Cleaning and handling missing values

Before performing analysis, the dataset was examined for inconsistencies, missing values, and structural issues. Data cleaning was carried out using Python (Pandas) to ensure accuracy and reliability of insights.

The dataset contains 3,900 records and 18 columns, including customer demographics, purchase details, subscription information, and review ratings.
From the initial inspection using df.info(), it was observed that:
Most columns contained complete data (3,900 non-null values).
The "Review Rating" column had 3,863 non-null values, indicating missing entries.
The missing values were found only in the Review Rating column.
Instead of replacing missing ratings with a global average, a more context-aware approach was applied:
The average review rating was calculated for each product category.
Missing review ratings were filled using the mean rating of the respective category.

```
[3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 350.5+ KB
```

4.Feature Engineering :

To perform better customer segmentation analysis, a new categorical column “Age Group” was created from the numerical Age column.
The age values were divided into four meaningful segments:

- Young Adult
- Adult
- Middle-aged
- Senior

This segmentation allowed more structured analysis of purchasing behavior across different demographic groups.

5.DataBase Integration:

After completing data cleaning and preprocessing in Python, the processed dataset was integrated into a MySQL database for structured storage and efficient data management.
The integration process included:

- Establishing connection between Python and MySQL using SQLAlchemy / PyMySQL
- Creating a structured database table
- Exporting the cleaned DataFrame into the MySQL database
- Verifying successful insertion of records

6.DataAnalysis using SQL

1.Revenue by Gender :

Revenue was analyzed across gender groups to understand their contribution to total sales. This helps identify the dominant customer segment and supports targeted marketing strategies.

gender	revenue
Filter...	Filter...
Male	157890
Female	75191

2. High Spending Discount Users (top 10)

High-Spending Discount Users – Customers who applied discounts yet recorded spending above the overall average purchase amount were identified. This segment highlights price-sensitive but high-value customers.

customer_id	total_spending
Filter...	Filter...
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

3.Shipping Type Comparison:

Purchase and revenue trends were compared across different shipping methods. This analysis provides insight into customer delivery preferences and operational efficiency.

shipping_type	revenue
Filter...	Filter...
Free Shipping	40777
Express	39067
Store Pickup	38931
Standard	38233
2-Day Shipping	38080
Next Day Air	37993

4. Top 5 Products by Rating:

   The top five highest-rated products were determined based on average customer reviews. This helps highlight customer preferences and maintain product quality standards.

item_purchased	Average Produc...
Filter...	Filter...
Gloves	3.861428571428573
Sandals	3.844375
Boots	3.818750000000001
Hat	3.801298701298701
Skirt	3.784810126582278

5. Subscribers vs Non-Subscribers :

   A comparison was made between subscribed and non-subscribed customers to evaluate differences in spending behavior. This helps measure the effectiveness of subscription programs.

subscription_sta...	revenue
Filter...	Filter...
No	170436
Yes	62645

6. Repeat Buyers & Subscriptions :

The relationship between repeat purchases and subscription status was analyzed. This helps evaluate whether subscription models improve customer retention.

subscription_sta...	repeat_buyers
Filter...	Filter...
Yes	958
No	2518

7. Revenue by Age Group :

Revenue was analyzed across different age groups to identify high-value demographic segments. This insight supports age-targeted marketing and product positioning strategies.

age_group	total_revenue
Filter...	Filter...
Young Adult	62143
Middle_aged	59197
Adult	55978
Senior	55763

8. Top 3 Products per Category :

The top three products within each category were identified based on performance metrics. This helps understand best-selling items within each segment.

item_rank	category	item_purchased	total_orders
Filter...	Filter...	Filter...	Filter...
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Customer Segmentation (Loyal, Returning, New)

Customers were segmented based on purchase frequency into loyal, returning, and new categories. This segmentation helps design targeted retention and engagement strategies.

customer_segm...	Number of Cust...
Filter...	Filter...
Loyal	3116
Returning	701
New	83

10 . Average Spending by Category

The average purchase amount was calculated for each product category. This analysis identifies high-value categories and supports pricing and inventory decisions.

category	average_spending
Filter...	Filter...
Footwear	60.2554
Clothing	60.0253
Accessories	59.8387
Outerwear	57.1728

7.DashBoard in Power Bi :

Interactive Power BI Dashboard Visualization

Business Recommendations

1.Focus on High-Revenue Categories

Since the Clothing category generates the highest revenue, the business should prioritize inventory expansion, targeted marketing campaigns, and premium product placement in this segment.

2.Target Young Adult Segment

Young Adults contribute the highest revenue. Personalized promotions, loyalty programs, and trend-based marketing strategies should be designed specifically for this age group.

3.Leverage High-Spending Discount Users

Customers who use discounts but still spend above average represent a high-value segment. Strategic promotional campaigns can be optimized to retain and upsell these customers.

4.Optimize Shipping Strategy

Analyzing shipping preferences allows the business to improve logistics efficiency and potentially promote preferred shipping methods to increase customer satisfaction.